

JAMES RUSE AGRICULTURAL HIGH SCHOOL
YEAR 12 MATHEMATICS EXTENSION II ASSESSMENT
TERM II 2004

QUESTION 1

Marks

The points $P(5p, \frac{5}{p})$ and $Q(5q, \frac{5}{q})$ lie on a rectangular hyperbola.

- | | | |
|-------|--|---|
| (a) | Graph the rectangular hyperbola clearly showing location of foci, asymptotes and directrices. Write the corresponding equations on the asymptotes and directrices. | 3 |
| (b) | Derive the equation of the tangent at P , hence state the equation of the tangent at Q . | 3 |
| (c) | The tangents from the points P and Q intersect at angle of 45° at the point T . | |
| (i) | Find the coordinates of the intersection point T . | 2 |
| (ii) | Show $(p^2q^2 + 1)^2 = (p^2 - q^2)^2$ | 2 |
| (iii) | Find expressions in terms of x and y : | |
| | (α) $p+q$ | 1 |
| | (β) pq | 1 |
| | (γ) $(p-q)^2$ | 1 |

Hence show that equation of the locus of T is given by :

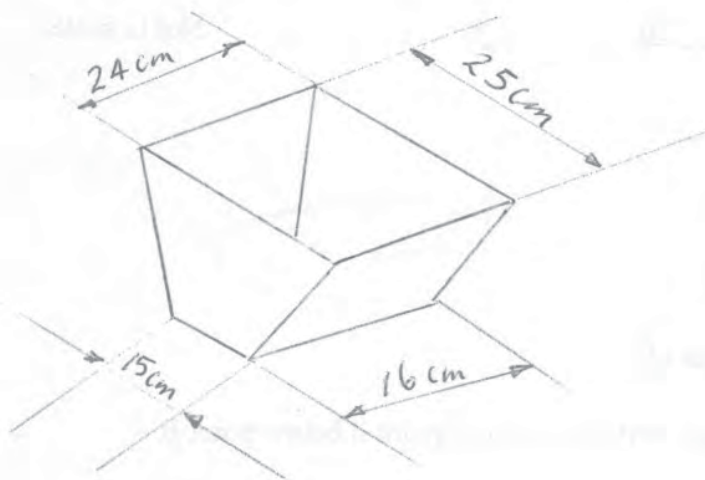
$$x^4 + y^4 + 2x^2y^2 + 400xy - 10000 = 0$$

(Neglect restrictions)

2

QUESTION 2 (START A NEW PAGE)

- | | | |
|-----|---|---|
| (a) | Find the volume of a symmetrical tray of height 5 cm if the cross sections parallel to the base are rectangles. | 5 |
|-----|---|---|



Not to scale.

- | | | |
|-----|---|---|
| (b) | The region bounded by the x axis, the curve $y = xe^x$ and the line $x=1$ is rotated about the y axis.
Using the method of cylindrical shells find the volume of revolution. | 5 |
| (c) | The region bounded by the coordinate axes and the curve $x^{\frac{1}{2}} + y^{\frac{1}{2}} = 1$ is rotated about the line $y = -1$. Find the volume of revolution. | 5 |

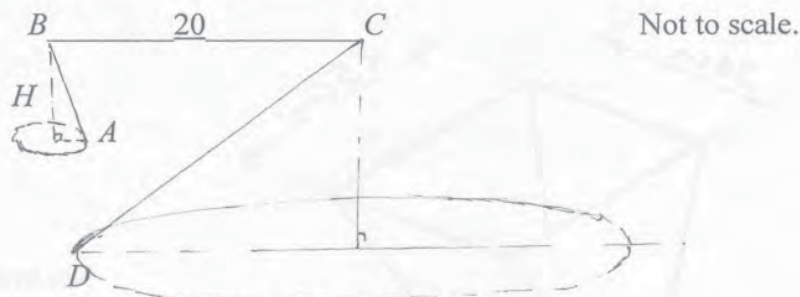
QUESTION 3 (START A NEW PAGE)

- (a)(i) A parachutist of mass 90 kg jumps out of an aeroplane at a height of 100 m.
The acceleration due to gravity is 10 m/s^2 .
Derive the vertical velocity functions $v \text{ m/s}$ in terms of displacement x for the parachutist, if the initial velocity is $u \text{ m/s}$ and there is:
- (α) no resistance force during free fall. 2
 - (β) a resistance force of $0.27v^2$ Newtons when the parachute is opened. 3
- (ii) The parachutist jumps from rest, and opens his parachute 60 m from the ground.
- (α) Find the vertical velocity when the parachute is opened. 1
 - (β) Find the vertical velocity (1 decimal place) on landing. 2
 - (γ) What percentage is the landing velocity compared to the terminal velocity ? 2
- (b) A particle moves in a straight line. 5
The displacement function x metres in terms of velocity $v \text{ m/s}$ is given by :
$$x = 5v - 2\ln(v+1)$$

Find the time t seconds of the particle as a function of it's velocity v , if the particle is initially moving at 1 m/s .

QUESTION 4 (START A NEW PAGE)

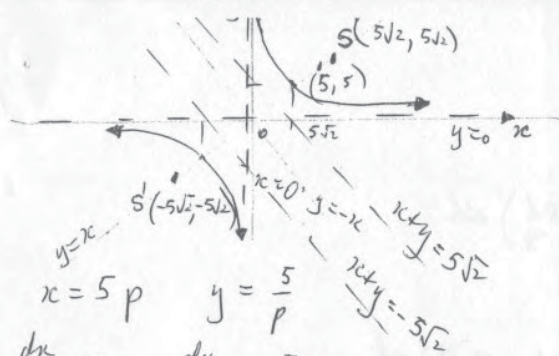
A light inextensible string AD passes over frictionless supports at B and C 20 cm apart.
At the ends of the string a mass M is attached to the end A 10 cm from B , and a mass m is attached to the end D 25 cm from C .
Both mass M and mass m rotate in the horizontal plane with uniform angular velocities of ω_1 and ω_2 respectively.
The acceleration due to gravity g is 9.8 m/s^2 .



- (a) Show a force diagram for mass M . 1
- (b) Prove $H\omega_1^2 = g$ where H is the vertical height of point A below point B . 4
- (c) Prove $m = \frac{2M\omega_1^2}{5\omega_2^2}$ 2
- (d) If $M = 200$ grams, $\omega_1 = 15 \text{ rad/s}$ then :
 - (i) find the vertical height H (nearest mm). 1
 - (ii) find the tension T Newtons in the string. 1
 - (iii) find the angular velocity ω_2 and corresponding mass m when the rotation of mass m comes into contact with the motion of mass M . 6

END OF EXAM

(c)



Focus $S(5\sqrt{2}, 5\sqrt{2})$
 $S'(-5\sqrt{2}, -5\sqrt{2})$
 Directrices $xy = 5$
 $xy = -5$

3.

$$(b) \quad x = 5p \quad y = \frac{5}{p}$$

$$\frac{dx}{dp} = 5 \quad \frac{dy}{dp} = -\frac{5}{p^2}$$

$$\frac{dy}{dx} = \frac{\frac{dy}{dp}}{\frac{dx}{dp}} = \frac{-\frac{5}{p^2}}{5} = -\frac{1}{p^2}$$

Eqn Tangent

$$y - y_1 = m(x - x_1)$$

$$y - \frac{5}{p} = -\frac{1}{p^2}(x - 5p)$$

$$py - 5p = -x + 5p$$

$$x + p^2y = 10p$$

$$\text{At } Q \quad x + p^2y = 10p$$

$$c(i) \quad y[p^2 - q^2] = 10[p - q]$$

$$y = \frac{10(p - q)}{(p - q)(p + q)}$$

$$y = \frac{10}{p + q} \quad p \neq q$$

$$x = \frac{10p}{p + q} - \frac{10p}{p + q}$$

$$= \frac{10pq}{p + q}$$

(1)

$$\Rightarrow T\left(\frac{10pq}{p+q}, \frac{10}{p+q}\right)$$

2

(ii) If intersect at 45°

$$\tan \theta = \left| \frac{m_1 - m_2}{m_1 + m_2} \right|$$

$$\tan 45^\circ = \left| \frac{\frac{-1}{q^2} + \frac{1}{p^2}}{1 + \frac{-1}{q^2} \cdot \frac{-1}{p^2}} \right|$$

$$1 = \left| \frac{p^2 - q^2}{1 + p^2q^2} \right|$$

$$\text{or } 1 = \frac{(p^2 - q^2)^2}{(1 + p^2q^2)^2}$$

$$\text{OR } (p^2q^2 + 1)^2 = (p^2 - q^2)^2$$

$$(iii) \quad y = \frac{10}{p+q} \Rightarrow p+q = \frac{10}{y}$$

$$\frac{x}{y} = \frac{10pq}{p+q} \div \frac{10}{p+q}$$

$$= pq$$

$$= pq = \frac{x}{y}$$

$$(p - q)^2 = (p + q)^2 - 4pq$$

$$= \left(\frac{10}{y}\right)^2 - \frac{4x}{y}$$

$$= \frac{100 - 4xy}{y^2}$$

(2)

$$\therefore (p^2 q^2 + 1)^2 = (p^2 - q^2)^2$$

$$(p^2 q^2 + 1)^2 = (p - q)^2 (p + q)^2$$

$$\left(\frac{x^2}{y^2} + 1\right)^2 = \frac{100 - 4xy}{y^2}, \left(\frac{10}{y}\right)^2$$

$$(x^2 + y^2)^2 = 10000 - 400xy$$

$$x^4 + y^4 + 2x^2y^2 + 400xy - 10000 = 0$$

(3)

Ques 2
(a)

$$h = 15 + 2h$$

$$h = 16 + \frac{8h}{5}$$

$$V = \int_0^5 (15 + 2h)(16 + \frac{8h}{5}) dh$$

$$= \int_0^5 (240 + 24h + 32h + \frac{16h^2}{5}) dh$$

$$= \left[240h + 28h^2 + \frac{16}{15}h^3 \right]_0^5$$

$$= 1200 + 700 + 133\frac{1}{3}$$

$$V = 2033\frac{1}{3} \text{ cm}^3$$

$$b) \delta V = 2\pi r, r e^k \delta r$$

$$\therefore V = 2\pi \int_0^1 r^2 e^k dr$$

$$= 2\pi \left[\left[r^2 e^k \right]_0^1 - \int_0^1 2r e^k dr \right]$$

$$= 2\pi \left[e - 0 - 2 \left(r e^k - e^k \right)_0^1 \right]$$

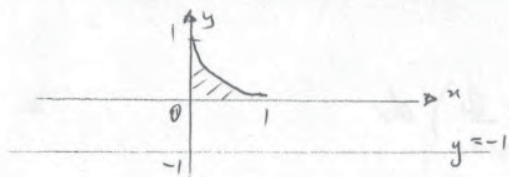
$$= 2\pi \left[e - 2 (e - e - (-1)) \right]$$

$$V = 2\pi [e - 2] \text{ unit}^3$$

(1)

(4)

(c)



$$y = (1-x^{1/2})^2$$

$$y = 1 - 2x^{1/2} + x$$

$$SV = \pi \int [(y+1)^2 - 1^2] \delta x$$

$$= \pi \int y(y+2) \delta x$$

$$V = \pi \int_0^1 (1 - 2x^{1/2} + x)(1 - 2x^{1/2} + x + 2) dx$$

$$= \pi \int_0^1 (1 - 2x^{1/2} + x)(3 - 2x^{1/2} + x) dx$$

$$= \pi \int_0^1 (3 - 2x^{1/2} + x - 6x^{1/2} + 4x - 2x^{3/2} + 3x - 2x^{3/2} + x^2) dx$$

$$= \pi \int_0^1 (3 - 8x^{1/2} + 8x - 4x^{3/2} + x^2) dx$$

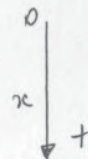
$$= \pi \left[3x - \frac{16}{3} x^{3/2} + 4x^2 - \frac{8}{5} x^{5/2} + \frac{x^3}{3} \right]_0^1$$

$$= \pi \left[3 - 5^{1/3} + 4 - 1^{3/5} + \frac{1}{3} \right]$$

$$V = \frac{2\pi}{5} \text{ unit}^3$$

Q.3

a(i)



$$(d) \quad m\ddot{x} = mg$$

$$\ddot{x} = 10$$

$$\int_u^v v dv = \int_0^x 10 dx$$

$$\frac{v^2 - u^2}{2} = 10x$$

$$v = \sqrt{u^2 + 20x} \quad v > 0$$

$$(b) \quad m\ddot{x} = mg - 0.27v^2$$

$$\ddot{x} = 10 - \frac{0.27}{90} v^2$$

$$\ddot{x} = 10 - 0.003v^2$$

$$\int_u^v \frac{v dv}{10 - 0.003v^2} = \int_0^x dx$$

$$\frac{-1}{0.003} \ln \left(\frac{10 - 0.003v^2}{10 - 0.003u^2} \right) = x$$

$$\ln \left(\frac{10 - 0.003v^2}{10 - 0.003u^2} \right) = -0.006x$$

$$\frac{10 - 0.003v^2}{10 - 0.003u^2} = e^{-0.006x}$$

$$10 - 0.003v^2 = (10 - 0.003u^2) e^{-0.006x}$$

$$v = \sqrt{\frac{10 - (10 - 0.003u^2) e^{-0.006x}}{0.003}} \quad v > 0$$

(5)

(6)

$$v = \sqrt{0^2 + 20 \times 40} \\ = 20\sqrt{2} \quad \text{OR } 28.28 \text{ m/s}$$

$$v = \sqrt{\frac{10 - (10 - 0.003 \times 800) e^{-0.008 \times 60}}{0.003}}$$

$$v = 39.6 \text{ m/s}$$

$$v_T = \sqrt{\frac{10}{0.003}} \\ = 57.7 \text{ m/s}$$

$$\% \frac{v_T}{v} = \frac{39.6}{57.7} \times \frac{100\%}{1} \\ = 68.6\%$$

$$x = 5v - 2 \ln(v+1)$$

$$\frac{dx}{dv} = 5 - \frac{2}{v+1} \\ = \frac{5v+3}{v+1}$$

$$x'' = v \frac{dv}{dx}$$

$$\frac{dv}{dt} = v \cdot \left(\frac{v+1}{5v+3} \right) \quad t=0 \quad v=0$$

$$\therefore \int \frac{5v+3}{v(v+1)} dv = \int dt$$

$$\text{But } \frac{5v+3}{v(v+1)} = \frac{a}{v} + \frac{b}{v+1} \Rightarrow 5v+3 = a(v+1) + bv$$

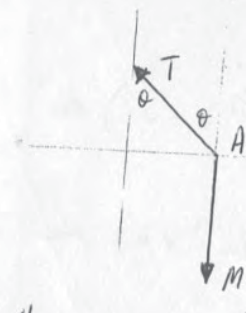
$$v=-1 \quad b=+2$$

$$t = \int_1^v \left(\frac{3}{v} + \frac{2}{v+1} \right) dv$$

$$= \left[3 \ln v + 2 \ln(v+1) \right]_1^v$$

$$t = 3 \ln v + 2 \ln \left(\frac{v+1}{2} \right)$$

Q.4



AB = 10 cm

Forces vertically

$$T \cos \theta = Mg$$

Forces horizontally

$$T \sin \theta = Mr \omega_1^2 \quad \text{And } r = 0.1 \sin \theta$$

$$= M \cdot (0.1 \sin \theta) \omega_1^2$$

$$T = 0.1 M \omega_1^2$$

$$\cos \theta = \frac{Mg}{0.1 M \omega_1^2}$$

$$\cos \theta = \frac{g}{0.1 \omega_1^2}$$

But

$$H = 0.1 \cos \theta \\ = 0.1 \cdot \frac{g}{0.1 \omega_1^2}$$

$$\Rightarrow H \omega_1^2 = g$$

(7)

(8)

(c) From (b) $T = 0.025 m \omega_2^2$ for mass m .

$$\begin{aligned} 0.25 m \omega_2^2 &= 0.1 M \omega_1^2 \\ m &= \frac{0.1 M \omega_1^2}{0.25 \omega_2^2} \\ &= \frac{2 M \omega_1^2}{5 \omega_2^2} \end{aligned}$$

(d) (i) $H = \frac{g}{\omega_1^2}$

$$= \frac{9.8}{15^2}$$

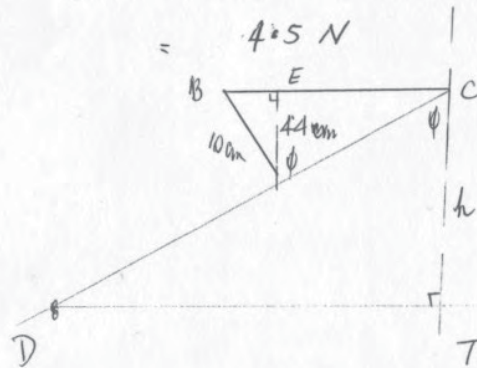
$$H = 44 \text{ mm}$$

(ii) $T = 0.1 M \omega_1^2$

$$= 0.1 \times 0.2 \times 15^2$$

$$= 4.5 \text{ N}$$

(iii)



$$BE = \sqrt{10^2 - 4.4^2}$$

$$BE = 8.98 \text{ cm}$$

$$CE = 20 - 8.98$$

$$CE = 11.02 \text{ cm}$$

$$\tan \phi = \frac{11.02}{4.4}$$

$$\phi = 68^\circ 14'$$

$$h = 25 \cos 68^\circ 14'$$

$$= 9.27 \text{ cm.}$$

$$\begin{aligned} h \omega_2^2 &= g \\ \omega_2 &= \sqrt{\frac{10}{0.0927}} \\ &= 10.38 \text{ rad/s} \end{aligned}$$

For mass from (c)

$$m = \frac{2 \times 0.2 \times 15^2}{5 \times 10.38^2}$$

$$= 1.67 \text{ gm.}$$

(9)